

BMW Car Sales Dashboard

Analysis & Findings Report

Executive Summary

This report summarizes the findings from the BMW Car Sales Power BI dashboard, which analyzes sales performance across various dimensions—time, model, color, region, and fuel type—using BMW vehicle transaction data. The aim was to identify key patterns and address important business questions for the sales or strategy team.

Below is a summary of the most important takeaways before we dive into the details.

Key Findings at a Glance

The dashboard tracks total revenue and units sold by car model, color, region, and fuel type. Users can interactively explore the data with four dynamic slicers, while KPI cards highlight the top-performing model, region, and year.

Dashboard Design & Structure

The report is designed on a single page to display everything at once, relying on filters for navigation.

1. KPI Summary Cards (Top Row)

When you open the report, the first thing you see is a row of four summary cards:

- Total Revenue: This shows the cumulative revenue from all selected data.
- Total Sales Volume: This represents the total number of BMW cars sold.
- Top Region: This card highlights the region that contributes the most revenue, along with its revenue figure.
- Top Model: This indicates the best-performing BMW model by revenue.

These cards are not just decorative; they are fully dynamic. Any changes you make with the slicers will update them instantly. For example, you can compare the top model in one region against another by adjusting the Region slicer.

2. Year-over-Year Combo Chart

The dashboard features a combined bar-and-line chart that displays annual performance. Bars show total units sold (Sales_Volume), while the line represents total revenue, revealing whether high sales volume aligns with revenue for the year.

3. Revenue by Car Color (Column Chart)

A simple column chart displays revenue by exterior color. This data informs production and inventory decisions, highlighting colors with unusually high or low revenue relative to their volume.

4. Revenue by Model (Treemap)

The treemap visually highlights model dominance, with larger tiles indicating higher revenue contributions. This format effectively showcases BMW's diverse model range, making it easy to identify the major revenue drivers like the 3 Series and X5 compared to niche models.

5. Revenue by Region (Donut Chart)

The regional donut chart shows revenue by geography. Clicking a slice cross-filters the dashboard to reveal popular models, colors, and fuel types in that market, aiding regional strategy discussions.

6. Revenue by Fuel Type (Donut Chart)

The fuel type breakdown is crucial in today's market, as automotive brands shift from combustion to electric and hybrid vehicles. The chart's tooltip indicates the number of cars sold per fuel type, revealing whether revenue share stems from volume or higher per-unit prices.

Analytical Observations & Findings

What the Data Dimensions Tell Us

Based on the structure and visuals in the dashboard, here are the analytical questions the report is designed to answer — and what the design implies about expected findings:

Revenue vs. Volume: They Don't Always Move Together

The combo chart highlights that in the automotive industry, revenue and unit volume can differ. When higher-margin models (like M-series or electric cars) sell fewer units but generate more revenue, the revenue line will rise faster than the bars. This indicates BMW's premium strategy of selling fewer but more expensive cars.

Model-Level Concentration Risk

The treemap highlights model-level revenue concentration. If a few models dominate the area, a slowdown in one could significantly impact overall performance due to dependencies.

Regional Imbalance

The regional donut chart shows BMW's geographic revenue concentration. A dominant slice indicates important business decisions for that region, while smaller slices highlight underpenetrated markets.

Fuel Type Transition

The fuel type breakdown is a key visual in the dashboard. If electric or hybrid vehicles are consistently increasing their revenue share, it indicates a market shift. The tooltip's car-count dimension also shows if EVs generate higher average revenue per unit.

Interactive Analysis Paths

One of the strengths of this dashboard is that it supports multiple analytical workflows depending on who is using it. Here are the most useful ways to interact with it:

Analyst Goal	How to Use the Dashboard
Track annual growth	Look at the combo chart across all years with no filters applied. The trend in both bars and line tells the headline story.
Compare regions	Use the Region slicer to toggle between regions and watch the KPI cards update. The top model and revenue figure will shift accordingly.
Understand EV vs. combustion mix	Select a recent year in the Year slicer and observe the fuel type donut. Compare with an earlier year to see the shift.
Find the best-selling color	Apply a model filter and look at the color column chart. Colors driving high revenue for a specific model can inform production.
Identify top model in a market	Select a region via the donut chart click or slicer, then read the Top Model KPI card.

DAX Measures & Calculations

The dashboard uses two calculated DAX measures on top of the raw columns. These are worth documenting because they're the backbone of every visual on the page.

Measure	What It Does	Used In
total revenue	SUM of Revenue column, respecting all active filters and slicer selections	All charts, KPI cards, tooltips
total sales volume	SUM of Sales_Volume column, respecting all active filters	KPI card, combo chart axis, fuel type tooltip

The Top Region and Top Model measures used in the KPI cards are dynamic text measures — they return the name (as a string) of the top-performing entry based on the current filter context, which is a nice touch for making the summary cards genuinely informative rather than just showing static numbers.

Potential Improvements & Next Steps

The dashboard is well-structured for a single-page overview. Here are some ideas for where it could go next, if the project were to evolve:

Add a Profitability Layer

Revenue tells part of the story. If margin or cost data were available, adding a profitability measure (revenue minus cost) would let the dashboard distinguish between high-revenue, low-margin models versus genuinely profitable ones — a crucial distinction for strategy teams.

Month-Level Granularity

The current time axis is year-level. If the data includes month or quarter, adding a date hierarchy to the combo chart would unlock seasonal analysis — for example, are there consistent Q4 spikes in certain regions or models?

Drillthrough Pages

Adding a dedicated model detail page (accessible via right-click drillthrough on the treemap) would let analysts go deep on a single model's performance across all dimensions without cluttering the main dashboard.

Target vs. Actual KPIs

If sales targets exist, adding a target line to the combo chart or a variance measure to the KPI cards would turn this from a descriptive dashboard into a performance management tool.

Geographic Map Visual

If the Region column maps to actual countries or territories, replacing the region donut with a filled map visual would make the geographic analysis more intuitive, especially for international stakeholders.

Conclusion

The BMW Car Sales dashboard is a streamlined single-page report that covers key automotive sales metrics and addresses common questions about revenue sources, leading models and regions, performance trends, and fuel type distribution through four interactive filters.

Designed for quick insights, the top KPI cards display key figures, while treemaps, donut charts, and combo charts provide complementary data views. Cross-filtering minimizes the need for data exports. The fuel type donut chart, especially combined with the Year slicer, reveals the shift towards electrification in BMW's revenue mix, which is crucial for long-term planning.